



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(15x2=30)

- I. What is importance of Raman Shift? Explain Scattered radiations behavior
- II. Consider the molecular vibrations of carbon dioxide and determine whether or not they are Raman active.
- III. Does the polarizability of each bond change in the bending vibration?
- IV. What are advantages of Raman spectroscopy on IR- spectroscopy?
- V. What do you know about molecular polarizability?
- VI. What is the principle of electronic transition?
- VII. What is the Franck Condon principle?
- VIII. What is meant by $n \rightarrow \pi^*$ transition?
- IX. Why ethanol is a good solvent in ultraviolet?
- X. What is the role chromophore in UV-Visible spectroscopy?
- XI. What type of transition is observed in methanol?
- XII. What is electronic angular momentum?
- XIII. What is vector summation for calculation of Angular momentum?
- XIV. What is compound doublet spectrum?
- XV. Raman spectra of asymmetric top molecules

Q.2. Answer the following questions.

(3x10=30)

- i) (a) Discuss rotational Raman spectra of linear molecules?
(b) Discuss Raman Effect with the help of an energy level diagram?
- ii) Discuss the energies of atomic orbital with reference to Hydrogen atomic spectrum?
- iii) Write note on following.
 - (a) Photoelectron spectroscopy
 - (b) Rotation-vibration Raman spectra of Symmetric top molecules



UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Eighth Semester –

Spring 2023

Roll No. ...

Paper: Physical Chemistry-II (UV & Raman Spectroscopy)

Course Code: CHEM-445

Time: 3 Hrs.

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Q.1. Answer the following short questions:

(15x2=30)

- ✓ i. What is Rayleigh scattering?
- ii. Why anti-stokes lines are less intense than stokes lines?
- ✓ iii. O_2 and Cl_2 are often found to be Raman active but inactive in IR. Justify?
- iv. What do you know about molecular polarizability?
- ✓ v. What are the spherical top molecules?
- vi. What is the principle of electronic transitions?
- vii. What is meant by $n \rightarrow \pi$ transition?
- ✓ viii. Why ethanol is a good solvent in ultraviolet?
- ix. What type of transition is observed in amines?
- x. What is the role of auxochrome?
- xi. What do you know about convergence limit in H-atomic spectrum?
- xii. What is electronic angular momentum?
- ✓ xiii. State Frank-Condon principle.
- ✓ xiv. What is vector summation for total angular momentum?
- ✓ xv. Write two applications of Raman Spectroscopy?

Q.2. Answer the following questions.

(3x10=30)

- ✓ A. Discuss rotational Raman spectra of symmetric top molecules? 450
- ✓ B. Explain five structure of H-atom spectrum 489
- C. Write note on the following?
 - i. Vibrational Raman spectra
 - ii. Photoelectron spectroscopy

UV Raman Spectroscopy (CHEM-445)

Short Questions:

- 1-What are the spherical top molecules? (2019)
- 2-What is meant by $n \rightarrow \pi^*$ transition? (2019)
- 3-What is the principle of electronic transition? (2019)
- 4-Define Raman Spectroscopy. (2019)
- 5-What is the role of molecular polarizability in Raman Spectroscopy? (2019)
- 6-What is the principle of electronic spectroscopy? (2020)
- 7-What are anti-stokes lines in Raman Spectroscopy? (2020)

Long Questions:

- 1-What is electronic spectroscopy? Write applications of electronic spectroscopy. (2019)
- 2-Write a note on the following: (2019)
 - (a) Vibrational-Raman Spectra
 - (b) Stark Effect
- 3-(a) What is Raman Spectroscopy? Discuss Rayleigh Scattering and molecular polarizability. (2020)
 - (b) Discuss pure Raman Spectra of linear molecules.
- 4-Discuss energy of atomic orbitals with reference to the H-atom spectrum. (2020)

M.Sc.-II(Past Papers)

- 1-What is the principle of electronic transitions? Give different types of electronic transitions. (2015, 2016, 2017, 2018, 2021)
- 2-What is Raman Spectroscopy? Discuss pure Raman spectra of linear molecules. (2016, 2017, 2018, 2019, 2020, 2021)
- 3-Write a note on the following: (2015, 2019)
 - (a) Stark Effect
 - (b) Rotational-Vibrational spectra